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Doppler Frequency Shifts Associated With the 18 September 2022 M6.8 Taitung Earthquake Observed by Local CW-HF Doppler Sounding Systems in Taiwan

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Key Points:

- Seismo-traveling atmospheric and ionospheric disturbances travel upward at about the speed of sound
- The horizontal speed of seismo-traveling ionospheric disturbances (STIDs) 1.4 km/s by nearby quakes is slower than that 3.2–3.9 km/s by faraway ones
- In the Taiwan vicinity, the total intensity of earthquakes is essential to trigger STIDs

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Doppler frequency shifts (DFSs) recorded by local continuous wave-high frequency (CW-HF) Doppler sounding systems have been used to study seismo-traveling ionospheric disturbances (STIDs) triggered by seismic waves of $M \geq 5.5$ earthquakes in the Taiwan vicinity during 2000–2022. We observed the first significant STID in DFSs induced by the 18 September 2022 M6.8 Taitung earthquake. To investigate this significant STID in more detail, DFSs of five sounding frequencies recorded by a network of nine receiving stations of the CW-HF Doppler sounding systems, together with the total electron content (TEC) derived by 35 ground-based GPS receivers, ionograms, and seismograms, are examined. Results show that horizontal speeds of the significant STID in DFSs of the 6.1 MHz sounding frequency and in TECs are 1.4 and 1.3 km/s, respectively. Upward propagation with a speed of sound of 0.72 km/s is estimated through DFSs of the five sounding frequencies, where the reflection heights are about 160–225 km altitude. We further examine DFSs associated with the top nine earthquakes in the Taiwan vicinity during 2022–2023. Results show that for near field observations, in addition to the earthquake magnitude, the total earthquake intensity plays an important role in triggering significant STIDs.

Plain Language Summary Previous scientists reported that seismo-traveling ionospheric disturbances (STIDs) in Doppler frequency shifts (DFSs) triggered by $M \geq 6.8$ earthquakes can be detected by continuous wave-high frequency (CW-HF) Doppler sounding systems at distances of thousands of kilometers away from the epicenters. They found that STIDs in DFSs travel horizontally radially away from the epicenter with speeds of 3.2–3.9 km/s. Here, DFSs recorded by local CW-HF Doppler sounding systems are used to observe STIDs triggered by seismic waves of 239 $M \geq 5.5$ earthquakes in the Taiwan vicinity during 2000–2022. On 18 September 2022, an M6.8 earthquake struck Taitung, and significant STIDs in DFSs induced by the earthquake were observed for the first time. DFSs of five sounding frequencies recorded by a network of nine receiving stations of local CW-HF Doppler sounding systems allow us to three-dimensionally examine STID propagation. We find that the upward speed of STIDs triggered by the 2022 M6.8 Taitung earthquake is the speed of sound, consistent with previous reports, but the horizontal speed is 1.4 km/s, which is lower than the previously reported 3.2–3.9 km/s. In addition to the earthquake magnitude, the total earthquake intensity plays an important role in triggering significant STIDs of DFSs in the Taiwan vicinity.

1. Introduction

An earthquake generates seismic waves that penetrate the Earth as body waves (P and S) or travel as surface waves (Love and Rayleigh). Rayleigh waves move in an elliptical motion, producing both a vertical and horizontal component of motion in the direction of wave propagation (Stein & Wysession, 2009). Vertical motions of the Earth's surface can efficiently trigger acoustic and/or gravity waves in the neutral atmosphere near the Earth's

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surface (hereafter termed, seismo-traveling atmospheric disturbances (STADs)), which propagate into the ionosphere and interact with the ionized gas, which is termed seismo-traveling ionospheric disturbances (STIDs) (Davies, 1990).

Davies and Baker (1965) might be the first to observe STIDs in continuous wave-high frequency (CW-HF) Doppler frequency shifts (DFSs) triggered by the 28 March 1964 M9.2 Alaskan earthquake, while Row (1967) suggested that the DFSs are caused by Rayleigh waves excited by the great Alaskan earthquake. Following that, many scientists using CW-HF Doppler sounding systems observe STIDs in DFSs triggered by seismic waves of large earthquakes, the 1968 M8.3 Hachinohe earthquake (Yuen et al., 1969); the 1982 M7.1 Urakawa-Oki earthquake (Tanaka et al., 1984); M6.8–8.3 earthquakes (Artru et al., 2004); the 2004 M9.3 Sumatra earthquake (Liu et al., 2006); the 2011 M9.0 Tohoku earthquake (Chum et al., 2012; Laštovička & Chum, 2017; Liu et al., 2016); the Nepal 2015 M7.8 earthquake (Chum et al., 2016; Laštovička & Chum, 2017); the M8.3 Illapel 2016 earthquake (Laštovička & Chum, 2017); the 2011 M7.3 Foreshock of the Tohoku earthquake (Nakata et al., 2021); the 2023 M7.8 Turkey earthquake (Salikhov et al., 2023). The above studies suggest that earthquakes greater than M6.8 may cause significant STIDs traveling with Rayleigh wave speeds of 3.2–3.9 km/s, which can be detected by Doppler sounding systems even thousands of kilometers away (far fields).

Liu et al. (2006) reported that STIDs in DFSs triggered by seismic waves of the 2004 M9.3 Sumatra earthquake were detected by Doppler sounding systems in Taiwan, which is about 3,600 km away from the epicenter, and the associated arrival seismic waves are about 0.02 cm (Fu & Sun, 2006). Similarly, Liu et al. (2016) reported DFSs triggered by seismic waves of the 11 March 2011 M9.0 Tohoku earthquake by CW-HF Doppler sounding systems in Taiwan, which is about 3,000 km away from the epicenter, and the associated arrival seismic waves are less than 5 cm (Hung & Rau, 2013). Due to the intense collision between the Philippine Sea and Eurasian plates, Taiwan has been frequently experiencing large earthquakes, which seems to be an ideal location to observe significant STIDs in DFSs induced by various magnitudes of earthquakes. In Taiwan, CW-HF Doppler sounding systems were established by the Telecommunication Laboratories, Ministry of Communication in 1981 (*cf*., Huang et al., 1985), and by the National Central University (NCU) in the late 1980s (Hung et al., 1990). The former stopped the observation after the devastating 1999 M7.6 Chi-Chi earthquake, while the latter has been continuously operating. Although STIDs in the total electron content (TEC) recorded by local ground-based GPS receiving stations were detected (Liu et al., 2010), the intense ground motion of seismic waves results in large areas' power off, and therefore, no DFS data were recorded by the two HF Doppler sounding systems during the 1999 M7.6 Chi-Chi earthquake. Based on the available data set, we have examined DFSs and 239 $M \geq 5.5$ earthquakes within a distance less than 400 km from the NCU CW-HF Doppler sounding system in Taiwan during January 2000–August 2022. The intensity within the distance of 400 km of the 239 earthquakes is large, and however, almost no obvious STIDS in DFSs can be observed during the 23-year period.

In this study, we report the first significant STIDs in DFSs triggered by an M6.8 earthquake that struck Taitung (Figure 1 and Figure S1), causing houses and bridges to collapse on 18 September 2022. The NCU CW-HF Doppler sounding system and the GPS receivers in Taiwan were employed to investigate the STID, including the horizontal and vertical propagation speeds. To have a better understanding of the causality of the significant STIDs, we also examine the total intensity of the top nine earthquakes in 2020–2023.

2. Observation

In Taiwan, NCU CW-HF Doppler sounding systems were established in the late 1980s and very recently renovated in 2020. The latest systems consist of a transmitting station with five sounding frequencies of 3.9, 4.4, 4.9, 5.5, and 6.1 MHz, and nine receiving stations at Taiwan and its outlying islands to three-dimensionally (3D) observe traveling ionospheric disturbances (TIDs) in DFSs triggered by solar flares, solar eclipses, magnetic storms, typhoons, earthquakes, etc. The network of receiving stations is used to compute the horizontal propagation and/or find the TID source location, while the five sounding frequencies recorded at each receiving station can be employed to estimate vertical propagation. Figure 1 illustrates the locations of transmitting and receiving stations, the reflection points of transmitted HF signals, as well as the epicenter of the 18 September 2022 M6.8 Taitung earthquake. The horizontal distances of the 45 ($=5 \times 9$) reflection points to the epicenter are less than 350 km (near fields).

Figure 2a illustrates the spectrum peak of DFSs (Liu et al., 2025), which shows a significant STID in DFSs at the sounding frequency of 6.1 MHz occurring at each station during the M6.8 Taitung earthquake. The STID arrival

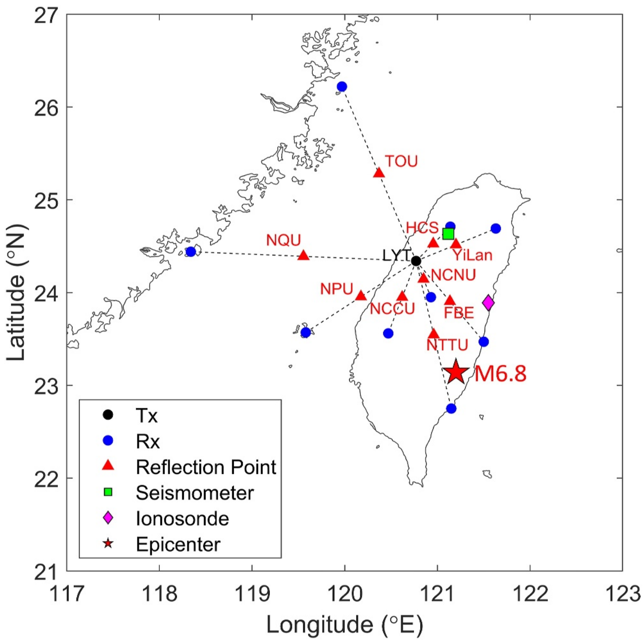

Figure 1. The M6.8 earthquake epicenter (red star) is located at 23.14°N, 121.20°E with a depth of 7.8 km on 18 September 2022. Locations of the HF Doppler Sounding System with transmitting station (black dot), receiving stations (blue dots), and their corresponding reflection points (red triangles). A seismometer and an ionosonde are denoted as green squares and magenta diamonds, respectively.

time of the nearest NTTU receiving station is 14:52LT (LT = UT+8 hr), which lags the earthquake occurrence at 14:44LT by about 8 min. This suggests that the travel time from the ground to the reflection point at 225 km altitude is somewhat slightly shorter than 8 min. The delay time increases as the distance to the epicenter increases, which indicates STIDs being observed. We apply the least-square fit on the nine detected STIDs, and find the associated horizontal speed of 1.38 km/s. The disturbed signals of STIDs generally agree well with the fitted line, except the STID of NTTU appearing earlier at about 14:52 LT.

Based on the method of intersecting circles (Lay & Wallace, 1995), which is traditionally employed for locating the hypocenter of an earthquake, we calculate the horizontal distance to a possible STID source location from each reflection point on the Earth's surface. The horizontal distance can be obtained by multiplying the horizontal speed by the horizontal propagation time, which is the time difference between the STID onset time and the earthquake occurrence time minus the vertical propagation time of 8 min. Thus, triangulation can be used to locate the STID source. The calculated horizontal distance from each reflection point to the STID source on the ground is shown as a circle. The location where most the circles intersect is the location of the STID source. Figure 2b shows that for STIDs in DFSs at the sounding frequency of 6.1 MHz, when the speed of 1.38 km/s is set, the distance between the epicenter and the intersection of circles (i.e., the computed epicenter) is the closest. It can be seen that seven reflection points of NTTU, FBE, NCCU, NCNU, NPU, YiLan, and HCS receiving stations are close to the Taiwan island, while those of NQU and TOU are over the Taiwan strait. Due to the larger deviation at the two ocean reflection points (NQU and TOU), the circle method simply employs the remaining seven land reflection points.

Figure 3a exhibits DFSs recorded at the YiLan receiving station and seismograms recorded by an NFF seismometer right under the reflection points of the sounding frequencies. Four out of the five DFSs depict clear STID signatures. The arrival times of 4.4, 4.9, 5.5, and 6.1 MHz at the reflection heights of about 160, 193, 212, and 225 km, which are obtained by applying POLAN (Liu et al., 1992; Titheridge, 1985) through local ionograms, are, respectively, at 14:52, 14:53, 14:53, and 14:52LT, while the arrival time of seismic waves in the NFF seismogram is 14:44LT. Thus, the average vertically upward velocities of STIDs estimated from the seismometer on the ground to the right above reflection points of the four DFSs are 0.33, 0.38, 0.39, and 0.45 km/s, and their overall average is about 0.39 km/s upward. Figure 3b displays the slope of the time between STID arrival and earthquake onset versus the four reflection heights at each station. The slope-fitting correlation coefficients at the six stations are generally greater than 0.7, except for the one over YiLan. On average, the upward speed in the ionosphere at 160–225 km altitude is 0.72 km/s.

Assume the ionospheric TEC to be on a thin shell about the Earth, located near about 350 km altitude (*cf.*, Jin et al., 2014; Liu et al., 2006, 2010). The intersection between the line of sight of a GPS satellite to a ground-based receiver and the shell is termed the ionospheric pierce point (IPP). Therefore, an IPP acts as a space seismometer recording TEC variation at 350 km altitude in the ionosphere. Here, the rate of changes in the TEC (rTEC) with the 1-Hz time resolution derived from 35 ground-based GPS receivers in Taiwan is employed to study STIDs triggered by seismic waves of the M6.8 earthquake. A high-pass filter of 400 s is applied on the rTEC. Figure 4a illustrates the locations of the ground-based receivers and IPPs of the four GPS satellites (PRN #15, #18, #23, and #24) around Taiwan during the M6.8 earthquake. Figure 4b depicts the rTEC on the distance of IPPs to the epicenter versus time. Significant STID signatures show that the horizontal speed of rTEC is about 1.30 km/s.

3. Discussion

Davies (1990) suggests that vertical motions of the Earth's surface, due to earthquakes creating mechanical disturbances, trigger STADs in the neutral atmosphere near the Earth's surface, which propagate into the ionosphere and interact with the ionized gas and become STIDs. Liu et al. (2016) employed concurrent/co-located

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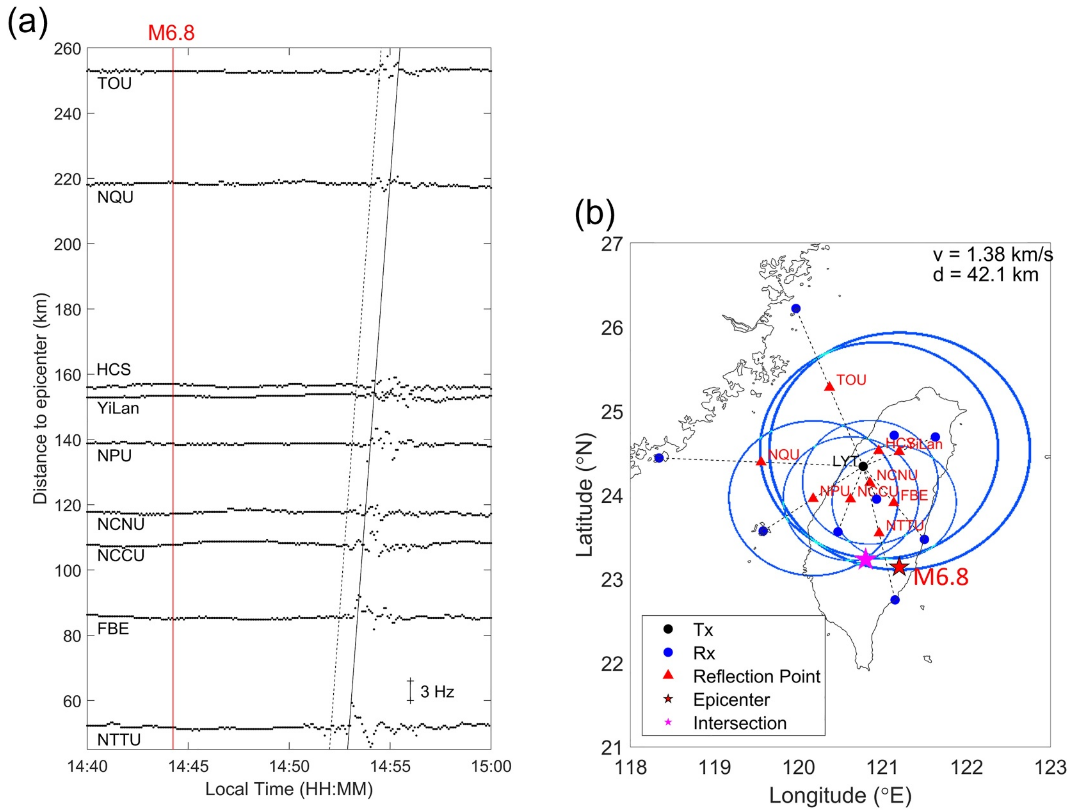

Figure 2. (a) The HF Doppler shift of 6.1 MHz during and after the M6.8 earthquake. The vertical red line registers the onset time of the earthquake, while the black solid and dashed lines indicate the speed of 1.38 km/s. (b) The circle method is based on the seismo-traveling ionospheric disturbances in Doppler frequency shifts at the sounding frequency of 6.1 MHz (including NTTU, FBE, NCCU, NCNU, NPU, YiLan, and HCS) with an estimated horizontal speed of 1.38 km/s. The magenta star indicates the intersection of circles, 42.1 km from the epicenter.

measurements of seismometers, infrasonic systems, magnetometers, Doppler sounding systems, and GPS receivers to study the propagation of STADs and STIDs triggered seismic waves of the 11 March 2011 M9.0 Tohoku earthquake. They found that the STADs and STIDs travel with the acoustic speeds in the atmosphere and ionosphere. Liu et al. (2019) used the GPS radio occultation of FORMOSAT-3/COSMIC (F3/C) with 1–2.5 km altitude resolution to scan fluctuations on vertical profiles of the ionospheric electron density induced by seismic and tsunami waves of the 11 March 2011 M9.0 Tohoku earthquake. They report that right underneath the Rayleigh waves can most efficiently induce significant STIDs traveling vertically upward. Liu et al. (2006, 2010, 2011, 2016) and Liu and Sun (2011) find that co-located seismic waves tend to lead the associated STIDs in rTECs or DFSs by about 8–10 min. These results show that the average vertical upward speed of STIDs from the ground to the reflection or IPP height is about 0.35–0.44 km/s, which agrees that the vertical upward speeds of STIDs estimated from the seismometer on the ground to the right above reflection points at 160–225 km altitude of the four DFSs are 0.33, 0.38, 0.39, and 0.45 km/s (Figure 3a). On the other hand, Figure 3b depicts the slopes of the arrival times of STIDs versus the reflection altitudes of the four Doppler sounding frequencies at the six selected stations. The slopes show that the overall upward speed of STIDs in the ionosphere at 160–225 km altitude is 0.72 km/s. This discrepancy could be due to the acoustic speeds being about 300 m/s from the ground to the E-region at 0–100 km altitude and varying from 300 to 880 m/s in the lower ionosphere at 100–225 km altitude (Liu et al., 2016). The speed of 0.39 m/s is obtained by averaging from the ground to the reflection heights of 150–225 km altitude, while the speed of 0.72 km/s is simply averaged over the reflection height ranges, which closes to the acoustic speed at about 200 km altitude (e.g., Liu et al., 2016). These results confirm that the STADs triggered by seismic waves near the Earth's surface travel vertically upward into the atmosphere and ionosphere with the speed of sound.

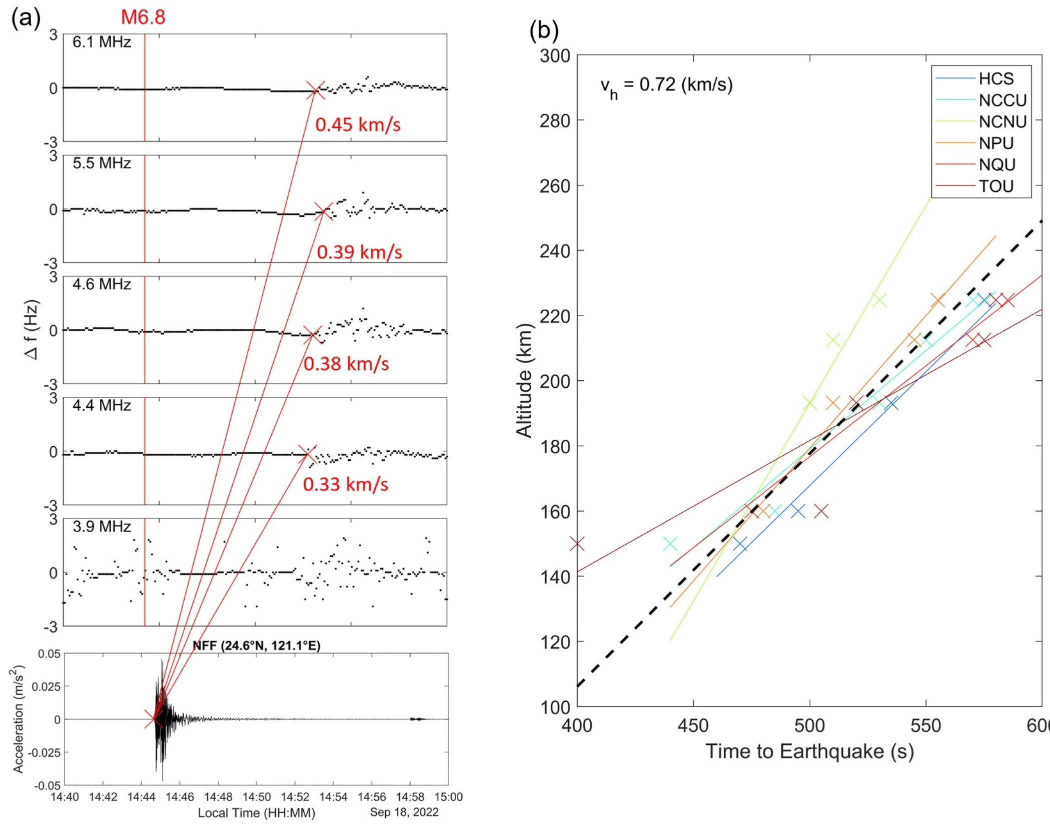

Figure 3. (a) From the top to low, the HF Doppler shift observed at YiLan with various frequencies from high to low. The bottom panel shows the seismometer (NFF) located around the reflection point of YiLan. The red crosses indicate the wave arrival time, while the estimated vertical average speeds are also listed. (b) The slope of the time between seismo-traveling ionospheric disturbance arrival and earthquake onset versus the four reflection heights at each station. The black dashed line shows an average speed of about 0.72 km/s.

The previous studies (Artru et al., 2004; Chum et al., 2012; Davies & Baker, 1965; Laštovička & Chum, 2017; Liu et al., 2006, 2016; Nakata et al., 2021; Salikhov et al., 2023; Tanaka et al., 1984) report that significant STIDs in DFSs caused by large earthquakes traveling with the horizontal speeds of 3.2–3.9 km/s which could be detected by remote Doppler sounding systems even thousands of kilometers away from the epicenter (i.e., a far field). In contrast, Liu et al. (2010) examined STIDs in rTECs derived by local ground-based GPS receivers in Taiwan during the 1999 M7.3 Chi-Chi-earthquake. They found that the horizontal speed triggered by the earthquake is about 1.7 km/s. Since their IPPs are located within a radius of about 500 km, they can be considered as a near field. On the other hand, Figure 4b shows that during the 2022 M6.8 Taitung Earthquake in Taiwan, the horizontal speed of STIDs in rTEC derived from the 35 local ground-based GPS receivers is about 1.30 km/s, which is similar to that of 1.38 km/s in DFSs recorded by the Doppler sounding system (Figure 2). The results of Liu et al. (2010) and Figures 2 and 4b show that the horizontal speed in near fields tends to be slower than that in far fields.

Among seismic waves, Rayleigh waves are often termed “ground rolls”, which are the most effective waves triggering STIDs. Telford et al. (1990) show that the typical speed of Rayleigh waves in the mantle is of the order of 2–5 km/s, and the typical Rayleigh speed on the ground is of the order of 50–300 m/s for shallow waves less than 100 m depth, and 1.5–4 km/s for depths greater than 1 km. It might be a mixture of Rayleigh waves with 50–300 m/s and 1.5–4 km/s close to the epicenter (i.e., the near field); the horizontal speed of STIDs in near fields tends to be slower than that in far fields. Meanwhile, owing to Rayleigh waves being confined near the surface, their in-plane amplitude when generated by a point source decays only as $1/\sqrt{r}$, where r is the radial distance. Surface waves therefore decay slowly with distance, which is one reason why Rayleigh waves of large earthquakes are measurably even at far fields of thousands of kilometers away from the epicenter. Meanwhile, for the Rayleigh speed of 3.2–3.9 km/s, the Mach number is about Ma8-11 in the atmosphere at 0–100 km altitude and

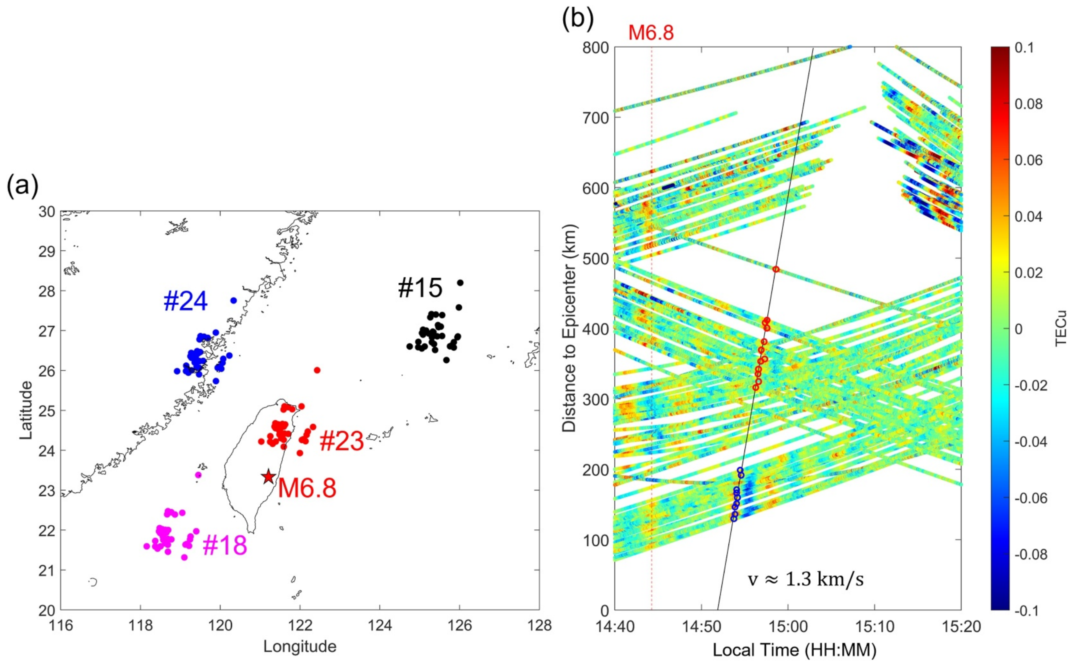

Figure 4. The GPS total electron content observations during and after the M6.8 Taitung earthquake. (a) The red, blue, black, and magenta dots show the ionospheric pierce points of each satellite associated with ground-based receivers. The red star indicates the epicenter of the earthquake. (b) The rTEC-time-distance plot from 35 ground-based GPS receivers in Taiwan. The rTEC is filtered by a high-pass filter of 400 s. The red and black lines denote the onset time of the earthquake and propagation speed of 1.3 km/s, respectively.

Ma2-3 in the bottom side of the ionosphere at 100–350 km altitude. Therefore, it should be the shock fronts, which effectively trigger STIDs in DFSs traveling with the Rayleigh speeds of 3.2–3.9 km/s reported by the previous scientists (Artru et al., 2004; Chum et al., 2012; Davies & Baker, 1965; Laštovička & Chum, 2017; Liu et al., 2006, 2016; Nakata et al., 2021; Salikhov et al., 2023; Tanaka et al., 1984).

Liu et al. (2006) examined giant STIDs in DSFs excited by the M9.3 Sumatra earthquake on 26 December 2004 by means of a network of Doppler sounding systems with three receiving stations in Taiwan. They found that an approximate horizontal speed from the epicenter to Taiwan is about 3.6 km/s in each DFS packet, which is almost the same as the Rayleigh waves in seismograms in Taiwan except with an 8-min time delay. On the other hand, a cross-correlation study (Liu et al., 1993) of the three arrival times of DFSs triggered by the Rayleigh waves shows an average horizontal speed of about 1.6 km/s coming from the epicenter (Liu et al., 2006). The discrepancy between the two might result from the former of 3.6 km/s is computed directly from the distance of the M9.3 Sumatra epicenter to Taiwan of about 3,600 km (i.e., far field) dividing the time delay of an individual DFS signal which is mainly contributed by the typical speed of Rayleigh waves in the mantle is of the order of 2–5 km/s, and the latter of 1.6 km/s is calculated by the local Doppler sounding system network, which simultaneously registers mixtures of Rayleigh waves of 50–300 m/s for shallow waves less than 100 m depth and 1.5–4 km/s at depths greater than 1 km. Therefore, it can be expected that a network might tend to underestimate the horizontal speed of STIDs because it observes mixtures of the primary and secondary waves.

In Taiwan, for far fields, upon the arrival of the 2004 M9.3 Sumatra earthquake and the 2011 M9.0 Tohoku earthquake, significant STIDs in DFSs triggered by underneath Rayleigh waves with the peak ground velocity (PGV) of 0.4 cm/s (Liu et al., 2006) and 0.01 cm/s (Liu et al., 2016) are observed. It is surprising to find that for near fields, almost no obvious STIDs in DFSs were triggered by 239 $M \geq 5.5$ earthquakes in Taiwan during 2000–2022 (Central Weather Administration Seismology Center, 1985), except for the 18 September 2022 M6.8 Taitung earthquake. To find possible causes of significant DFSs triggered by the 2022 M6.8 Taitung earthquake, we further examined the intensity of the top nine earthquakes with magnitude M5.8–6.8 during 2020–2023. Based on CWASIS (Central Weather Administration seismic intensity scale), Intensity 1 to 7 corresponds to PGV less than 0.2 to greater than 140 cm/s (Central Weather Administration Seismology Center, 1985). Figures 5a–5h and 5j

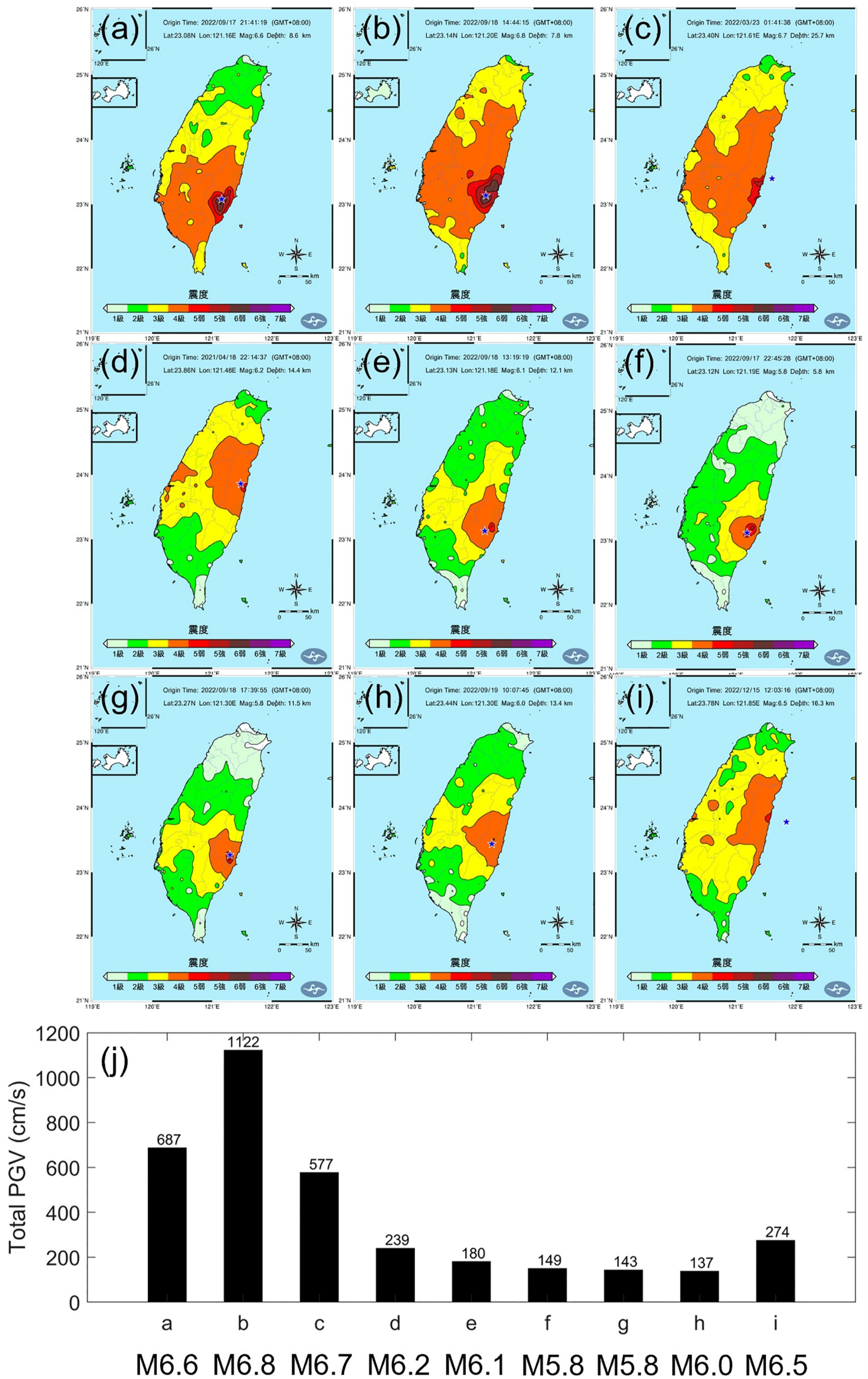

Figure 5. The intensity maps of the top 9 strongest earthquakes during 2020–2023 (a–i). The 18 September 2022 M6.8 Taitung earthquake is displayed in (b). The lowest panel (j) displays the total peak ground velocity of each earthquake (https://scweb.cwa.gov.tw/zh-tw/earthquake/data/).

display the spatial distribution of Intensity and the PGV sum (i.e., total earthquake intensity) of the nine earthquakes, respectively. The PGV sum of 1,122 cm/s (about 9.0 cm/s for the average PGV) of the 2022 M6.8 Taitung Earthquake is much greater than that of the other earthquakes. This suggests that the total PGV plays an important role. By contrast, the 2004 M9.3 Sumatra earthquake with a PGV of 0.4 cm/s and the 2011 M9.0 Tohoku earthquake with a PGV of 0.01 cm/s are much smaller than that of the largest intensity of the nine earthquakes, which vary from intensity 5 to 7 (i.e., PGV: about 30 cm/s to greater than 140 cm/s). Note that the two PGVs are registered at a single point in Taiwan. It should be noticed that for far fields, although the PGV is small, the coherent (resonant) and supersonic of seismic waves can efficiently activate STADs near the Earth's surface and STIDs at the right above.

Figure 2a shows that the STID of NTTU appeared at about 14:52 LT, which is earlier than the time estimated by the list square fit. It can be seen that NTTU is the nearest reflection point from the epicenter and close to the edge of the large PGV area (Figure 5b). Thus, NTTU experiences an area-like seismic wave source, while the rest experience a point-like one. Since an area source has a better amplification efficiency than a point source, the STIDs of NTTU appear at an earlier time of about 14:52 LT.

4. Summary and Conclusion

Previous studies report that Doppler sounding systems can detect STIDs in DFSs triggered by $M \geq 6.8$ earthquakes at long distances. These far field STIDs travel with the speed of sound in the upward direction and with 3.2–3.9 km/s in the horizontal direction. In the current near field study, the local Doppler sounding systems are used to find STID signatures possibly associated with 239 $M \geq 5.5$ local earthquakes in Taiwan during 2000–2022. The latest Taiwan Doppler sounding systems with five sounding frequencies recorded by nine receiving stations for the first time register STIDs in DFSs triggered by the 2022 M6.8 Taitung earthquake in Taiwan. The DFSs of the five sounding frequencies show that STIDs travel upward with the speed of sound. The DFSs recorded by nine receiving stations in the near field area display that the horizontal STID speeds are about of 1.4 km/s, which are much slower than those triggered by the far field earthquakes of 3.2–3.9 km/s. The study on the top nine earthquakes during 2020–2023 shows that the total intensity is essential to trigger observable STIDs in DFSs in near fields.

Data Availability Statement

The NCU Doppler Sounding System data used for analyzing ionospheric variations in the study are available at Zenodo via https://doi.org/10.5281/zenodo.14841350 with no license and other access restrictions (Liu et al., 2025). Data sets for this research are available in these in-text data citation references: Central Weather Administration Seismology Center (1985) [with no license needed]. Such data sets of historical seismic data and the intensity map of 18 September 2022 must be findable and accessible (e.g., via https://scweb.cwa.gov.tw/en-us/earthquake/data and https://scweb.cwa.gov.tw/en-us/earthquake/ShakeMap/EE2022091814441568111).

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